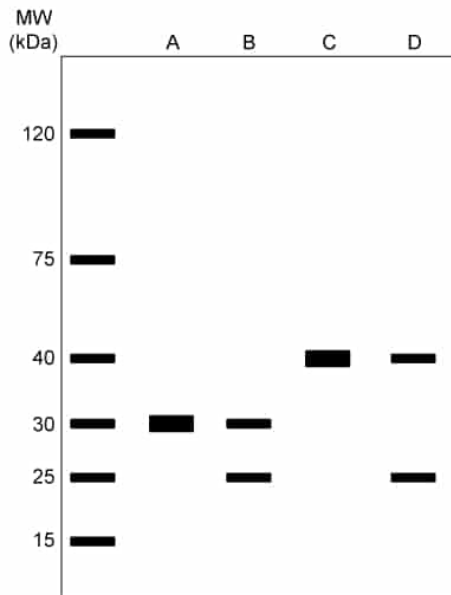


Proteins A, B, C, and D are dimers. They were run on a reducing SDS gel, shown below:



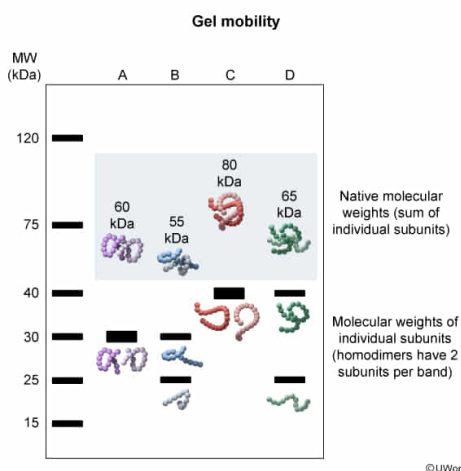
What is the order of molecular weights of the proteins in their native state, from largest to smallest?

- ☐ A. Protein D → Protein C → Protein B → Protein A [4%]
- ☒ B. Protein C → Protein D → Protein A → Protein B [54%]
- ☐ C. Protein D → Protein B → Protein C → Protein A [31%]
- ☐ D. Protein C → Protein A → Protein D → Protein B [8%]

Correct

55% answered correctly

Explanation:



In their native (ie, properly folded) form, many proteins consist of **multiple polypeptide chains** known as **subunits**. These proteins, called **multimers**, may consist of two subunits (**dimers**), three subunits (**trimers**), four subunits (**tetramers**), or more.

When a multimer runs through a **reducing SDS gel**, the sodium dodecyl sulfate (SDS) causes proteins to unfold, and the reducing agent breaks any **disulfide bonds** that may link subunits together. Each subunit migrates through the gel by itself, and **subunits of different sizes are separated**. For multimers in which all subunits are identical (**homomultimers**), all subunits migrate the same distance and appear as a **single band**. If the subunits are not identical (**heteromultimers**), they migrate different distances according to molecular weight and appear as **separate bands**.

According to the question, **each protein in the gel is a dimer**. Lanes A and C each show single bands, indicating that these proteins are **homodimers**. In contrast, Proteins B and D separate into two distinct bands, indicating that they are **heterodimers**. In their native (biologically active) form, the dimers have a molecular weight equal to the **sum of the weights of their subunits**. For heterodimers, the weights of each band (indicated by the marker lane) can be added. For homodimers, the weight of the single band can be doubled to account for both subunits. From largest to smallest, the order of the proteins is **Protein C (80 kDa) → Protein D (65 kDa) → Protein A (60 kDa) → Protein B (55 kDa)**.

(Choices A and C) Protein C is the largest of the proteins shown at 80 kDa and should be the first in order, followed by D, A, and B.

(Choice D) Proteins A and D each have a 25-kDa subunit. However, protein D also has a 40-kDa subunit, whereas Protein A has a 30-kDa subunit. Therefore, protein D is larger than protein A.

Educational objective:

In their native form, many proteins consist of multiple polypeptide chains (subunits) and are known as multimers (eg, dimers, trimers, tetramers). Reducing SDS gels separate multimer subunits by size. Identical subunits migrate the same distance and form a single band, whereas distinct subunits migrate different distances and form separate bands.

Time spent:0 sec

Subject:
Biochemistry

QID:402513

System:
Amino Acids and Proteins

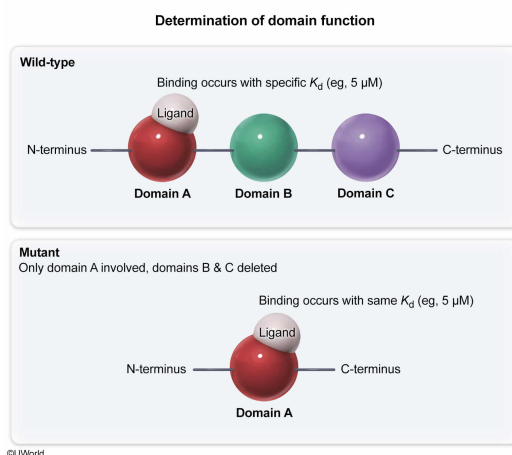
A protein consisting of domains A, B, and C binds a ligand. Which experimental result would provide the best evidence that only domain A is involved in binding and that domains B and C do not contribute to binding?

- ✔ ☒ A. A mutant containing only domain A binds the ligand with the same K_d as the wild-type protein. [41%]
- ☐ B. A mutant containing only domains B and C binds the ligand with a lower K_d than the wild-type protein. [3%]
- ☐ C. A mutant containing only domain A binds the ligand with a greater K_d than the wild-type protein. [6%]
- ☐ D. A mutant containing only domains B and C cannot bind the ligand. [48%]

Correct

41% answered correctly

Explanation:



Protein domains are **distinct regions** in a polypeptide chain that **fold independently**. In general, each domain carries out a separate function, although one domain may assist another in performing its function. One of the most common functions of a protein domain is to **bind a small molecule, called a ligand**. The affinity of a protein for a ligand is measured by the **dissociation constant K_d** , where a **small K_d** value corresponds to **high binding affinity** (or a low tendency of the complex to dissociate).

In this question, the protein consists of domains A, B, and C. If domain A is the *only* domain involved in ligand binding, then **binding affinity should be unaffected** in the absence of domains B and C. Therefore, if a mutant that contains only domain A can bind the ligand with the **same K_d as the wild-type protein**, this suggests that **only domain A is involved in ligand binding** and the other domains are **not involved**. In contrast, if domains B or C were involved in binding, removing them would affect the value of K_d .

(Choice B) A decreased K_d value corresponds to a *higher* binding affinity, so if domains B and C alone have a *lower* K_d for the ligand than the wild-type protein, this would suggest that **domain A interferes** with ligand binding.

(Choice C) An increased K_d value corresponds to a *lower* binding affinity. If domain A alone has a *higher* K_d for the ligand than wild-type binding, this would suggest that domain B, domain C, or both *aid in binding*.

(Choice D) If a mutant that contains only domains B and C cannot bind the ligand, this shows that domain A is *required* for binding. However, this does not rule out the possibility that domains B or C *may also contribute by helping A to bind* (eg, Choice C) or even that these domains are *required themselves*. For example, if domains A and B are *both* required for binding, removing *either one* would result in no binding. Consequently, removing only domain A does not fully address the role (if any) of domains B and C in binding.

Educational objective:

Protein domains are distinct, independently folding regions of polypeptide chains that generally carry out distinct functions. Ligand binding by a protein domain is measured by the dissociation constant K_d . A high value of K_d corresponds to low affinity for the ligand, and vice versa.

Time spent:0 sec

Subject:
Biochemistry

QID:402514

System:
Amino Acids and Proteins